

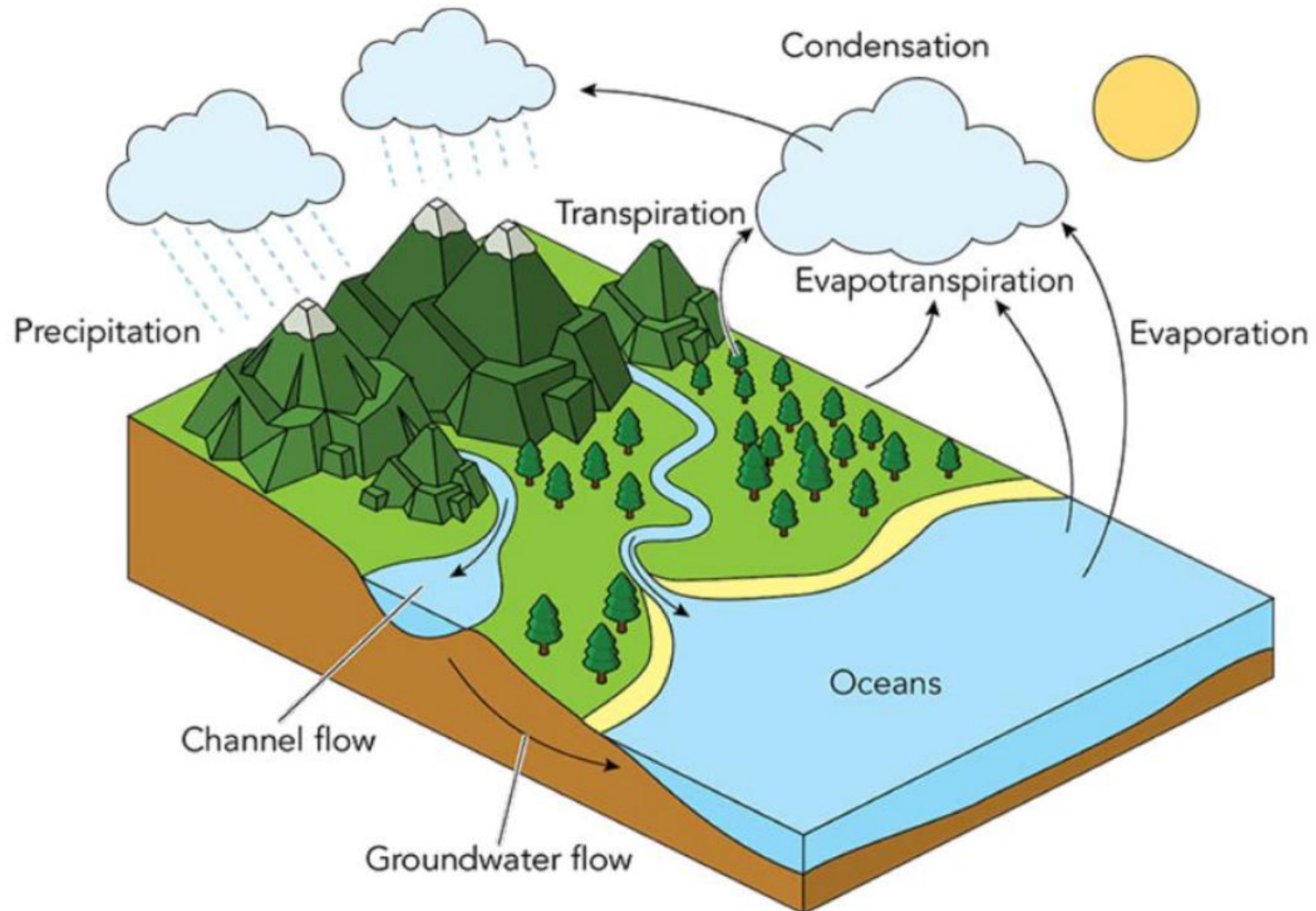
1.1 Rivers and drainage basins

The water cycle

The water cycle, or hydrological cycle, refers to the movement of water between the atmosphere, land and sea. It is a cycle because there is no start or end.

Water falling as **precipitation** either:

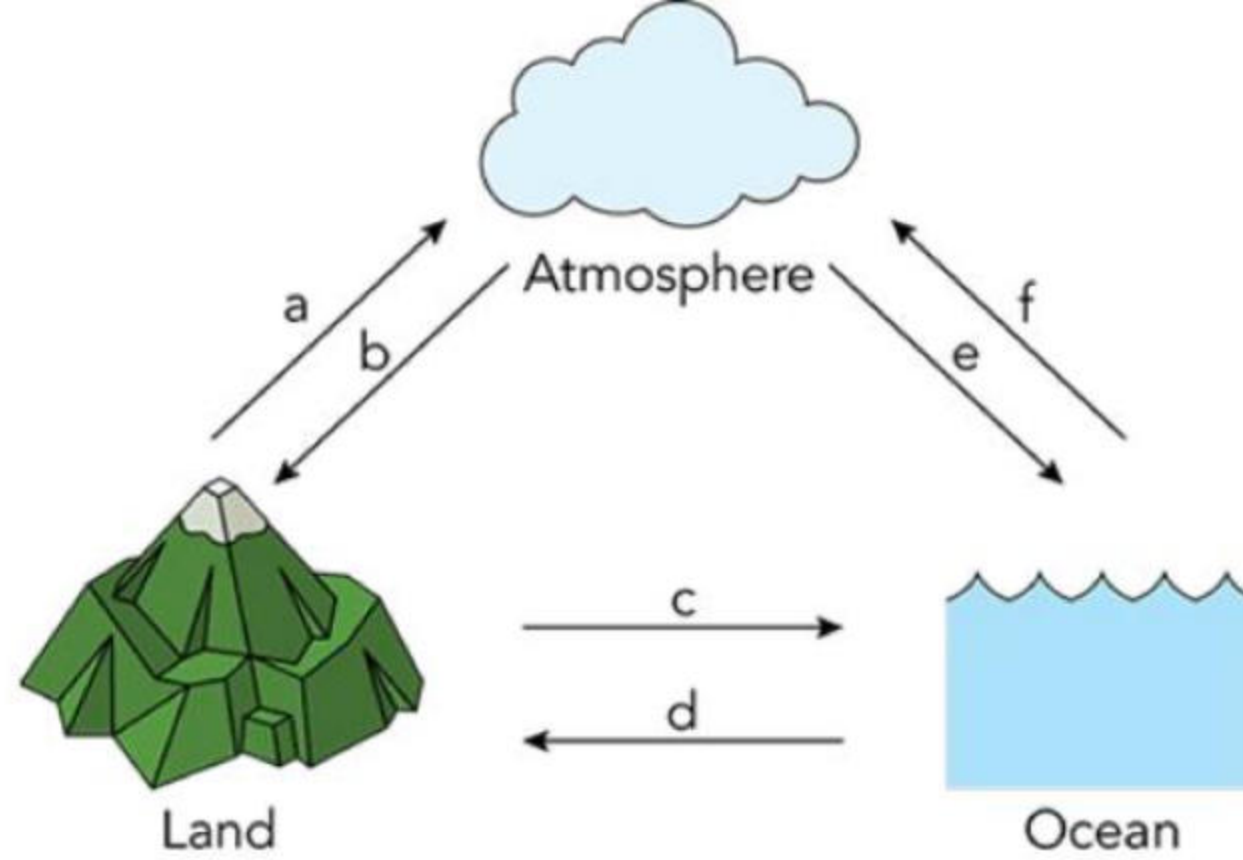
- flows over the land as channel flow
- flows through the land as groundwater
- travels back into the atmosphere through evapotranspiration
- is stored on or within the land, for example, in vegetation, soils, lakes, or glaciers.



Source **A**: The water cycle

TASK 1: Study Source **A**

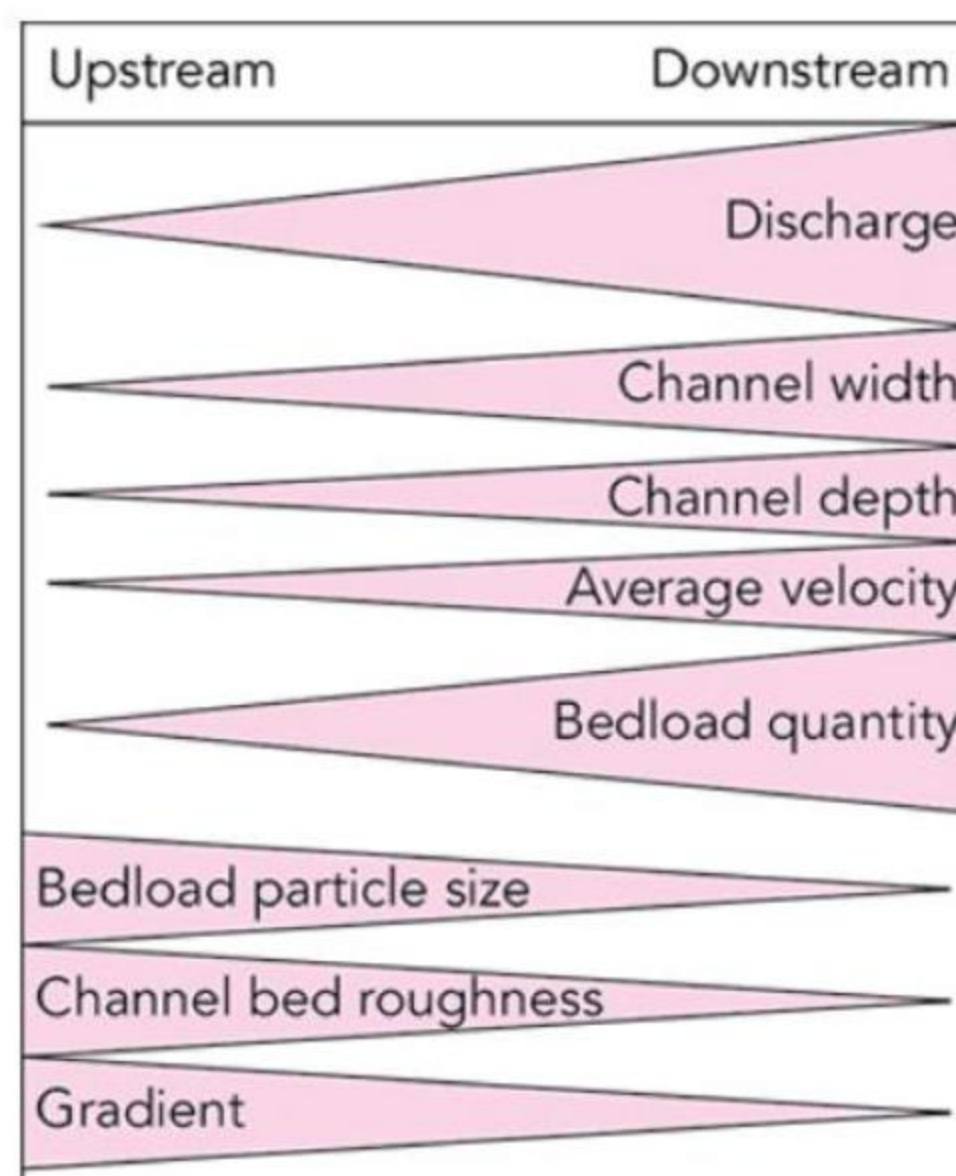
- Make a copy of the diagram below. Label arrows a, b, c, e, and f to show how water moves between the atmosphere, ocean, and land.
- Look at the diagram below. Explain why water does not move directly from the ocean to the land (arrow d).



Rivers

Rivers are an important part of the water cycle. They are channels of water that flow over the Earth's surface. The start of a river is called the **source** and the end is called the **mouth**. Through gravity, rivers transport precipitation that falls onto the land down towards the sea.

The characteristics of rivers change between the source and the mouth. The Bradshaw model is a theoretical model that shows these changes. In the model, triangles are used to represent changes along the course of a river.



Source **B**: The Bradshaw model

In upland areas, closer to the source, rivers tend to be narrow and shallow. They have a smaller **wetted perimeter** and a lower **discharge**, because they carry less water than lowland rivers. Upland rivers have a lower velocity, which means they are slower flowing. This is because the large bedload (material located on the riverbed) creates friction, which slows the water down.

In lowland areas, closer to the mouth, rivers are wide and deep. They have a much larger wetted perimeter and a higher discharge because they have flowed over a longer distance, collecting more water as they go. Lowland rivers have a higher velocity (speed of flow), as there is less friction from the riverbed and banks.



Source **C**: An upland river



Source **D**: A lowland river

TASK 2: Study Sources **B** **C** **D**

Create a **table** like the one below. Complete all boxes in the table by choosing the correct word from each cell.

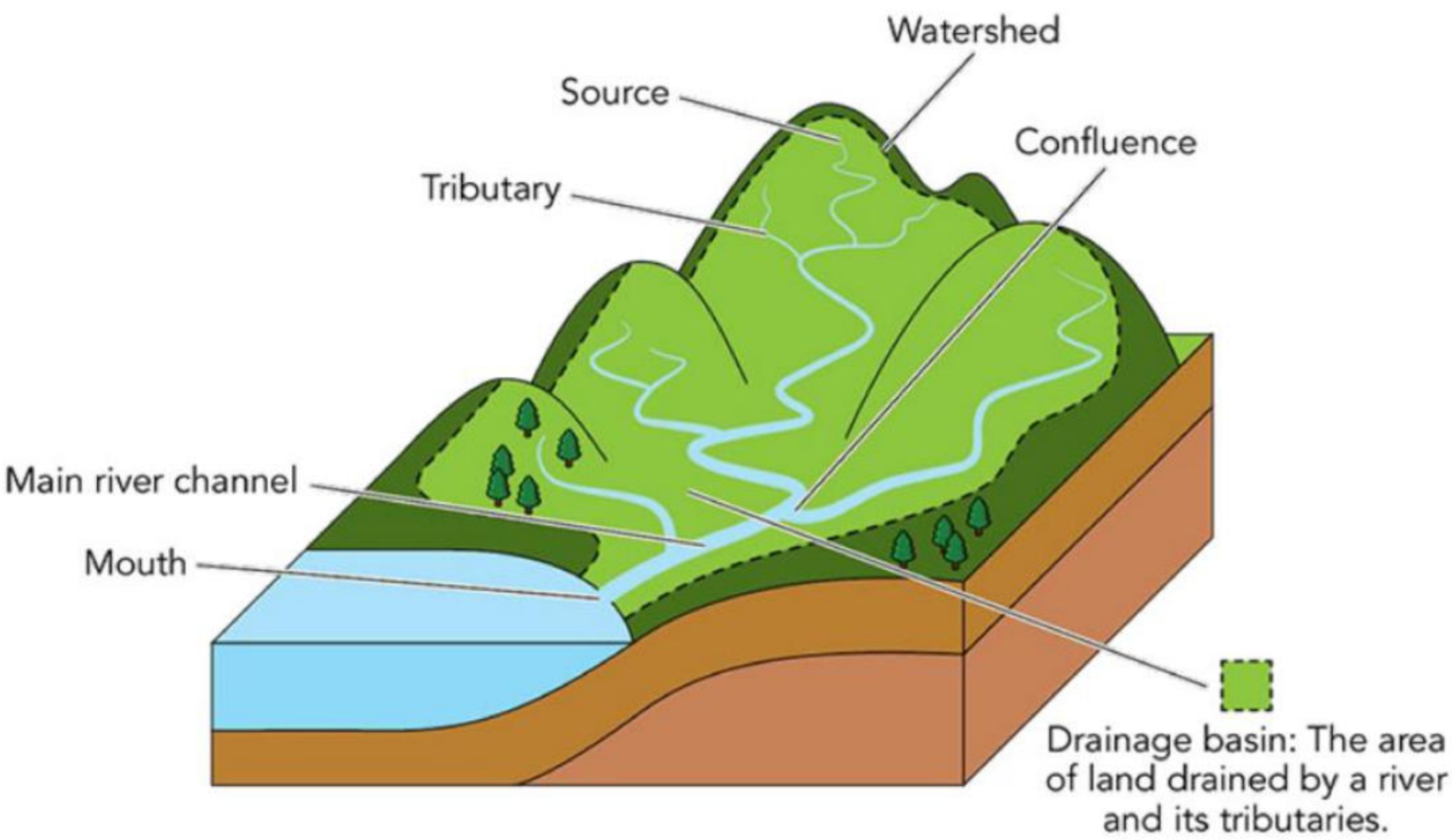
	River source	River mouth
Width	wide/narrow	wide/narrow
Depth	deep/shallow	deep/shallow
Velocity	faster/slower	faster/slower
Discharge	higher/lower	higher/lower
Gradient	steep/gentle	steep/gentle

TIP

It is easy to get confused about changes in river velocity between the source and the mouth. Think of it like a ball rolling down a hill; the ball will speed up as it rolls and will be travelling much faster at the bottom of the hill.

Drainage basins

Rivers flow within **drainage basins**. The drainage basin is an important part of the hydrological cycle, moving water from upland areas towards the sea. The edge of a drainage basin is marked by the watershed, which is usually found on higher ground. This acts as a dividing line, where precipitation falling one side of the watershed flows into one drainage basin and water falling the other side flows into another. Most of the Earth’s surface is covered in a network of connecting



Source **E**: A drainage basin

The river within a drainage basin rarely follows a single path. Smaller streams, or tributaries, feed into the main river channel. A confluence is found where two main channels meet.

Several processes operate within a drainage basin which determine the flow of water through the system. These include:

Table 1.1: Processes operating in the drainage basin

Process	Description
Precipitation	water that moves from the atmosphere to the land, e.g. rain and snow
Interception	vegetation prevents precipitation from reaching the ground
Infiltration	surface water soaks into the surface of the soil
Percolation	water moves through the soil and rocks
Throughflow	water from the soil moves into rivers and streams
Groundwater flow	water stored deep underground moves into rivers and streams
Evaporation	water stored on the surface moves back into the atmosphere
Transpiration	intercepted water moves back into the atmosphere
Evapotranspiration	water moves back into the atmosphere by evaporation and transpiration
Overland flow	water runs over the surface and into rivers and streams
Channel flow	water is transported by rivers and streams

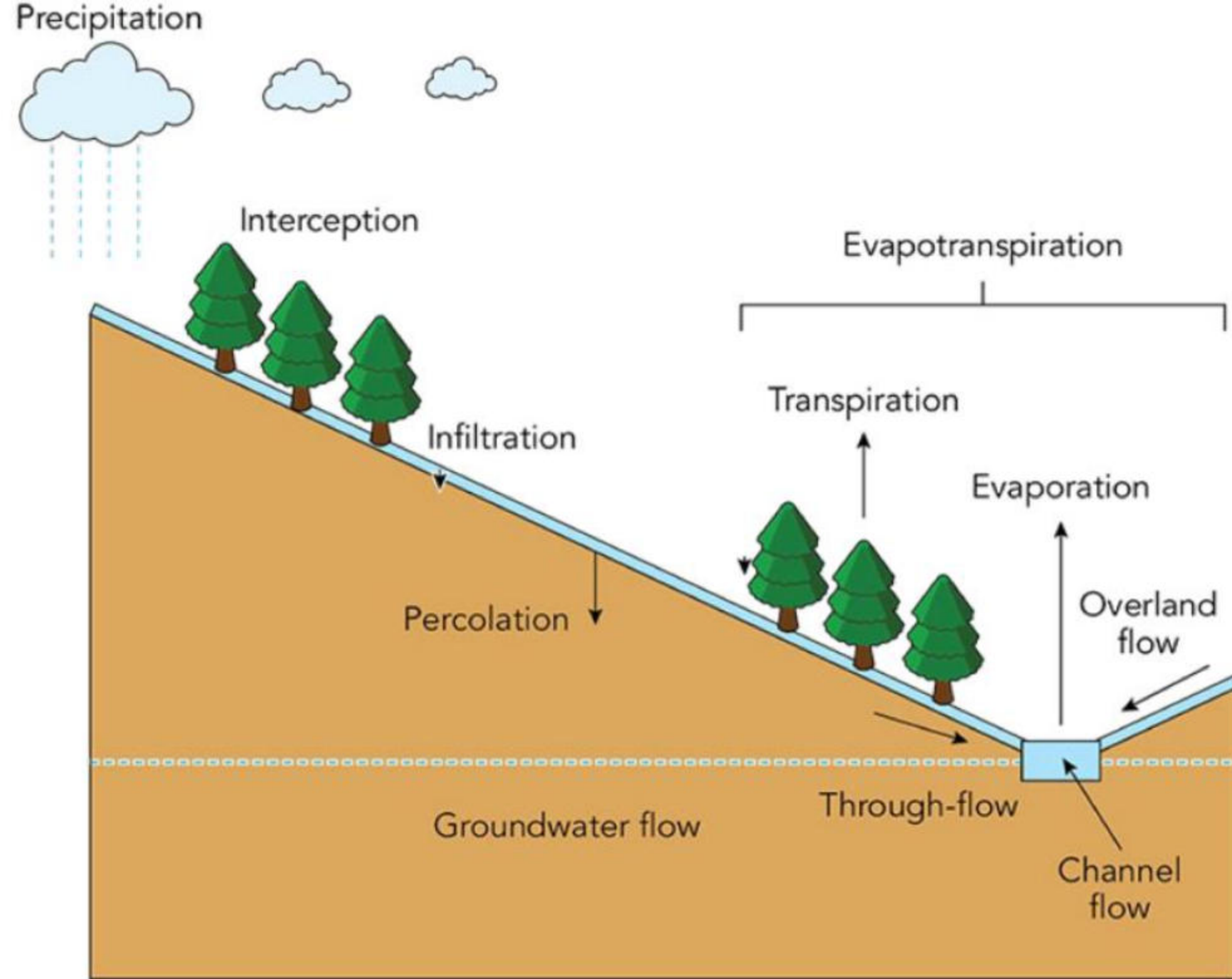
TASK 3: Study Source **E**

a

Explain why the watershed is found on higher ground.

b

Explain why river discharge increases between the source and the mouth.



Source F: Processes that operate in a drainage basin

An understanding of rivers and drainage basins is important for managing water resources, preventing flooding, and preserving ecosystems.

REFLECTION

The differences in a river between the source and the mouth are easier to remember once you understand why the changes take place. Think about why rivers become wider and deeper, and why the velocity and the discharge increase. Does understanding how the processes work help you to remember the differences between the different parts of a river? Do you think this approach will help you in other geographical topics?

TASK 4: Study Source F

Drainage basins are open systems, which means that water can enter or leave the basin.

- Explain how water enters the drainage basin system.
- Explain how water leaves the drainage basin system.

TASK 5

Work in pairs. Imagine you were travelling in a boat from the source of the river to the mouth. Describe your journey. Think about things like the landscape around you and the speed at which you're travelling. Once you've finished writing your journey, join up with another pair from the class and share your stories. Peer-assess one another's work.